

Precision-AMR: Spatially Adaptive Floating-Point Precision for PDE-Based Reverse Time Migration

*A new axis of numerical adaptivity for adjoint wave-equation solvers,
developed on dedicated NVIDIA hardware and validated across
heterogeneous NVIDIA and AMD GPU architectures on the Santos
Dumont supercomputer (LNCC)*

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1 Problem Statement

GPU hardware is increasingly shifting toward low precision in favor of AI workloads (FP8/BF16), exposing new memory hierarchies, tensor cores, and mixed-precision math libraries. At the same time, scientific simulations based on partial differential equations (PDEs) continue to require FP64/FP32 precision to remain trustworthy. Reverse Time Migration (RTM), a seismic imaging method that solves the wave equation by finite differences over thousands of time steps, is a canonical example of this tension: geophysicists use the migrated image to guide expensive drilling decisions, so numerical fidelity is essential. RTM is also memory-intensive, requiring hundreds of gigabytes of wavefield state and relying on checkpointing to trade memory against recomputation in its forward/backward adjoint structure.

Our prior work tackled this structure's main bottleneck, namely host–GPU checkpoint transfer over PCIe, by combining the concept of prefetching with data compression. This

idea was materialized as the GPUZIP library [4, 5], which compresses checkpoints on the GPU and prefetches them proactively, reducing host–GPU transfer volume by up to $80\times$ and achieving end-to-end speedups of up to $5.1\times$ on V100, while preserving image quality ($\text{SSIM} \approx 1$, near-zero relative error). Separately, reduced-precision arithmetic has already been shown to be viable for seismic modeling, RTM, and FWI: Fabien-Ouellet [2] found that a stable half-precision (FP16) implementation of the elastic wave equation is achievable through appropriate scaling, that the resulting seismogram error is a negligible fraction of total seismic energy and largely incoherent with seismic phases, and that this noise does not adversely affect FWI or RTM image quality. However, that reduction is applied uniformly across the entire computational domain. Gao and Harms [3] identified a specific hazard of that approach: naive half-precision time stepping accumulates round-off error through a disguised recursive summation in the update rule, degrading solution quality, with compensated (Kahan) summation as an effective remedy.

At the same time, adaptive mesh refinement (AMR) has been explored for seismic wave simulation and, more recently, explicitly for FWI and RTM workflows, to avoid the waste inherent in a uniform grid: dispersion and dissipation errors require grid spacing proportional to the shortest wavelength, so a uniform mesh is forced to use the finest spacing everywhere, even where the local velocity field would tolerate a much coarser one. AMR addresses this by refining Δx locally. No comparable, physically motivated scheme exists for floating-point precision, however: existing mixed-precision work in seismic imaging is either global and uniform (as above), or adaptive only at the level of an individual linear-algebra operation (mixed-precision GEMM, iterative refinement, Ozaki-scheme emulation), not tied to a spatial decomposition of the wavefield itself. None of this prior work varies floating-point precision according to physically motivated spatial criteria. This is therefore a third axis of adaptivity that remains largely unexplored.

This project proposes to close that gap with Precision-AMR (P-AMR): a spatially block-structured scheme, an adaptive-precision analogue of AMR, that keeps mesh resolution fixed and instead varies the floating-point format assigned to each block (tile) of the domain, based on physically motivated error-tolerance criteria (proximity to absorbing boundaries, wavefield energy density, adjoint-sensitivity magnitude, and a time-dependent band trailing the propagating wavefront) that predict whether the round-off error introduced in that block can reach the imaging condition or the FWI gradient. This is a structurally different problem from mesh-based AMR. In AMR, the characteristic artifact is a localized reflection at the coarse-fine grid interface, because refinement changes the discretization itself. In P-AMR, no discretization boundary is introduced; instead, a spatially varying level of round-off noise is injected into a hyperbolic PDE, and that noise propagates together with the wave. Tagging criteria therefore cannot rely solely on local smoothness, as AMR refinement indicators do; they must instead approximate where an error, once injected, will decay, be absorbed, or otherwise fail to reach the receivers or the sensitivity kernel

before the simulation ends, and control round-off accumulation over the thousands of time steps typical of RTM/FWI.

Because RTM/FWI already require a block/tile decomposition of the domain for checkpointing, P-AMR reuses that same decomposition as its tagging granularity, unifying the precision budget and the lossy-compression budget of differential checkpointing into a single, jointly tuned per-tile accuracy target. Mixed-precision GEMM, cuSOLVER iterative refinement, and Ozaki-scheme FP64 emulation remain essential as supporting techniques, but are now applied selectively, only where P-AMR’s criteria indicate they are needed. This is the natural way to exploit GPUs geared primarily toward AI, such as the RTX 5090, which offer negligible native FP64 throughput.

Because the tagging criteria are motivated by the physics of wave propagation and adjoint sensitivity, rather than by any particular vendor’s arithmetic units, we treat their portability across architectures as an explicit question, not an assumption. Our research group has access to the Santos Dumont supercomputer (LNCC/SINAPAD), which includes AMD Instinct MI300A accelerators alongside several NVIDIA generations (Volta V100, Hopper H100, Grace-Hopper GH200). This lets us test that question directly, rather than merely assert it, and it is what gives Precision-AMR its potential contribution to the field: a numerical adaptivity axis that is new and portable across vendors, not merely a one-off optimization for a single GPU architecture.

2 Expected Results

The project’s principal contribution is the Precision-AMR (P-AMR) method itself, which explores a new axis of numerical adaptivity that varies floating-point precision spatially, according to physically motivated criteria (proximity to the PML, wavefield energy density, adjoint-sensitivity magnitude, and a wavefront-trailing band), proposed and validated specifically for adjoint PDE solvers such as RTM/FWI. This contribution includes the formulation of the four tagging criteria, the characterization of the “precision-interface” artifact, analogous to the coarse-fine interface reflections of classical AMR, and the demonstration that the precision and lossy-compression budgets can be unified into a single per-tile error target.

As secondary contributions, the project will produce: (i) a multi-architecture evaluation of P-AMR-guided mixed-precision usage, comparing NVIDIA GPUs (V100, H100, GH200) and the AMD Instinct MI300A APU, characterizing how much of the achieved speedup and accuracy trade-off is vendor-independent; and (ii) the implementation of P-AMR as an **independent library**, with an interface that allows its integration into other PDE-based simulation applications, beyond the specific scope of RTM/FWI. This library may be adopted standalone or combined with GPUZIP, merging P-AMR’s spatial precision zoning with GPUZIP’s compressed checkpointing into a single workflow, without requiring the host

application to adopt the entire GPUZIP stack just to benefit from P-AMR. It will be made available as open source, under the MIT license, on GitHub.

The central scientific result will be submitted to a high-impact high-performance-computing event or journal, for example, Supercomputing (SC), IPDPS, Euro-Par, or SBAC-PAD among the events, or IJHPCA or a Transactions-type journal from IEEE or ACM among the journals, accompanied by a public reproducibility package (datasets, Nsight profiles, quality metrics), made available via UNICAMP's REDU.

3 Scientific and Technological Challenges and the Means and Methods to Overcome Them

The feasibility of adaptive mesh refinement (AMR) for reducing the computational cost of seismic simulation is already well established in the literature. van Driel et al. [8] and Thrastarson et al. [6] show, respectively, that wavefield-adapted meshes accelerate forward/adjoint modelling and FWI by up to an order of magnitude when medium properties vary smoothly. Yang et al. [9] apply a variable-spacing mesh specifically to elastic RTM, preserving multicomponent imaging accuracy while reducing computational cost and storage. Davis and LeVeque [1] further show that refinement indicators guided by the adjoint computation, and not just by local smoothness of the field, produce more efficient mesh refinements in hyperbolic wave-propagation problems, the same logic that motivates P-AMR's sensitivity (FWI) criterion. None of these works, however, varies *floating-point precision* instead of *mesh resolution*. Nor does any of them characterize empirically how the numerical behavior of an RTM/FWI simulation, that is, the dynamic range of floating-point values actually exercised, distributes across space and time. This double gap, the absence of a physically motivated precision-zoning scheme and the absence of the empirical characterization needed to ground it before designing the criteria, defines this project's scientific and technological challenges, organized in the six phases below, over a 36-month programme.

3.1 Phase 1 — Infrastructure and Baseline (M1–M4)

Challenge: before any precision change is attempted, it is necessary to faithfully reproduce the GPUZIP/Awave-3D baselines under the HPC SDK and ensure that the block decomposition used for checkpointing can also carry a per-block precision tag, an engineering prerequisite that every subsequent phase depends on.

Method: port Awave-3D and GPUZIP to the HPC SDK (nvc++), re-integrate nvCOMP/Bitcomp, and reproduce the V100 baselines on Santos Dumont's V100 partition, matching the original GPUZIP hardware.

- Establish a block/tile decomposition of the domain, extending the existing checkpoint-block structure so that it can also carry a precision tag per block.
- Profile every kernel with Nsight Systems and Nsight Compute, on both the RTX 5090 and Santos Dumont, confirming that FDM stencils are memory-bandwidth-bound at the tile granularity before any precision change is attempted.

3.2 Phase 2 — Empirical Characterization of Mixed-Precision Behavior (M4–M10)

Challenge: the four tagging criteria proposed for Phase 3 are physically motivated, but, unlike the adjoint-guided mesh refinement of Davis and LeVeque [1], no work in the literature empirically characterizes how the dynamic range of floating-point values (exponents and magnitudes actually used) behaves spatiotemporally in an RTM/FWI simulation, nor investigates whether that behavior can be anticipated directly from the input velocity model, before the wave even propagates. Without that characterization, the tagging criteria remain plausible hypotheses that have not yet been verified against actual numerical behavior.

Method: instrument the solver to record, from FP64 reference runs, dynamic-range statistics of the wavefield per block and per time step: exponent histogram, minimum/maximum magnitude, variance, across the three datasets (Large, Marmousi3D, Salt).

- Correlate these statistics with attributes derived from the input velocity model (local impedance contrast, depth, distance to reflectors, local wavelength), seeking which attributes are predictive of where the dynamic range, and therefore the tolerance to round-off error, is structurally lower.
- From that correlation, build an *a priori* precision-zoning estimator, computable directly from the velocity model before any time step is run. This estimator is complementary to the four reactive criteria of Phase 3, which depend on the wavefield computed during the simulation itself.
- Use this characterization to validate, or refute, the physical assumptions behind each of the four proposed criteria, before investing in their implementation and fine-tuning in Phase 3.

3.3 Phase 3 — Precision-AMR: Criteria, Tagging, and Validation (M10–M20)

Challenge: unlike mesh-based AMR, where the characteristic artifact is a localized reflection at the coarse-fine interface, round-off noise induced by precision reduction propagates together with the wave itself, so tagging criteria cannot rely solely on local smoothness of the field, as classical refinement indicators do. They must anticipate where an error, once injected, will fail to reach the receivers or the FWI gradient before the simulation ends. A

second, technological challenge compounds this: reduced precision and lossy compression, both already used separately in GPUZIP, may interact non-trivially when applied to the same block, and that interaction needs to be isolated before defining a joint error budget.

Method: this is the core scientific phase, run at scale on Santos Dumont’s H100 partition. Building on the characterization from Phase 2, we design, implement, and validate four candidate tagging criteria for spatial precision zoning:

- **Boundary criterion:** tiles inside or adjacent to the PML/absorbing layer are tagged for reduced precision by default, since their contribution is already being damped.
- **Energy criterion:** tiles are tagged by a threshold on local wavefield energy density, so that regions of low amplitude (before wavefront arrival, or after geometrical spreading has attenuated the signal) tolerate lower precision.
- **Sensitivity criterion (FWI):** tiles are tagged using the magnitude of the local adjoint-sensitivity kernel, reusing the gradient computation itself as the tagging signal, in direct analogy to the adjoint-guided refinement of Davis and LeVeque [1] and to the Berger–Oliger truncation-error estimator in classical AMR.
- **Wavefront-trailing criterion:** a time-dependent band immediately behind the propagating wavefront is tagged for reduced precision, while the leading edge, where phase accuracy governs the imaging condition, retains full precision.

For each criterion, we implement compensated (Kahan) summation in the low-precision tiles’ time-stepping update to control round-off accumulation over the full simulation, following the mechanism identified by Gao and Harms [3]. We explicitly test for a “precision-interface” artifact, analogous to coarse-fine interface reflections in mesh-based AMR, via residual analysis at the boundaries between tiles of differing precisions.

To isolate the effects of mixed precision from the effects of lossy compression before unifying them into a single per-tile error budget, each candidate tagging region is evaluated under four conditions, always compared against the FP64 reference by PSNR, SSIM, and mean relative error, on both the raw wavefield and the final migrated image:

1. full precision, no compression (control);
2. full precision, with differential compression (isolates the compression effect);
3. reduced precision, no compression (isolates the precision effect);
4. reduced precision, with differential compression (the condition actually deployed in production).

Comparing conditions (ii)+(iii) against (iv) indicates whether the two effects are approximately additive, whether one masks the other, or whether there is amplification through

interaction. Only then does each tile jointly receive a floating-point format and a checkpoint compression policy (full snapshot versus compressed delta), as a single, jointly tuned error budget. This phase subsumes differential checkpointing, which becomes the storage mechanism for P-AMR-tagged tiles rather than a separate deliverable.

3.4 Phase 4 — Precision-AMR without Native FP64 (M19–M27)

Challenge: GPUs geared primarily toward AI, such as the RTX 5090, offer negligible native FP64 throughput. Emulating FP64 across the entire domain via the Ozaki scheme [7] is technically viable but expensive enough to make indiscriminate use impractical, which only reinforces the need to apply it selectively.

Method: the tagging criteria validated in Phase 3 are applied on the RTX 5090. Per tile, the pipeline selects between native low-precision tensor-core execution, cuBLAS mixed-precision GEMM (FP32 compute with FP64 accumulation) or cuSOLVER iterative refinement for dense sub-kernels, and Ozaki-scheme FP64 emulation for tiles flagged as accuracy-critical.

- This directly operationalizes the motivation for exploiting AI-first GPUs that lack native FP64: expensive emulation is concentrated only where P-AMR’s criteria indicate it is needed.
- We quantify the speedup/accuracy trade-off relative to a uniform-Ozaki baseline (emulating FP64 everywhere) to determine how much selective tiling actually saves.
- All errors are bounded jointly (compression $\times$ precision interaction, following the Phase 3 methodology) against the FP64 baseline using PSNR, SSIM, and mean relative error, across the same three datasets and three checkpointing algorithms (Revolve, zCut, Uniform) used throughout the project.

3.5 Phase 5 — Cross-Architecture Validation (AMD Instinct MI300A) (M26–M31)

Challenge: the tagging criteria are motivated by the physics of wave propagation and adjoint sensitivity, not by any particular vendor’s arithmetic units. That portability, however, is a claim that needs to be tested, not merely assumed, especially because AMD offers no direct equivalent to the Ozaki scheme or to nvCOMP/Bitcomp.

Method: with the dedicated time that a 36-month programme allows, we port the same tagging and tiling logic to Santos Dumont’s AMD Instinct MI300A nodes via ROCm/HIP (rocBLAS, hipBLAS, and AMD’s own reduced-precision matrix paths), without assuming feature parity with the CUDA-specific path.

- We test whether the Phase 3 criteria and the precision $\times$ compression factorial methodology hold on a CDNA3 architecture, re-running the same evaluation protocol used on the NVIDIA GPUs.

- We test whether the *a priori* estimator built in Phase 2, derived from the velocity model and architecture-independent, transfers directly, without retuning, to the MI300A.
- We compare the speedup/accuracy trade-off achieved on the MI300A against the NVIDIA baselines, to characterize how much of the gain is architecture-independent and how much is specific to the CUDA/Ozaki/nvCOMP stack.

3.6 Phase 6 — Spectral Operators, Integration, and Release (M31–M36)

Challenge: consolidating the supporting techniques (differential checkpointing, mixed-precision GEMM, Ozaki-scheme emulation, spectral operators) and the Precision-AMR module into a single, coherent, reusable software artifact, the P-AMR library described in Expected Results, rather than leaving them as isolated prototypes from each phase.

Method:

- Benchmark cuFFT-based pseudo-spectral derivative operators as an alternative update rule for selected, precision-tagged tiles, as a secondary technique complementing the core P-AMR contribution.
- Integrate the Precision-AMR module, already as an independent library, differential checkpointing, and mixed-precision paths into a unified GPUZIP v3.0 release.
- Multi-node scaling evaluation on Santos Dumont (≤ 8 GPUs).
- Final analysis, write-up, and submission, with Precision-AMR as the central scientific claim and differential checkpointing, mixed-precision linear algebra, and spectral operators presented as supporting techniques.

3.7 Computational Platforms

HPC SDK (nvc++/nvfortran, CUDA-X math libraries, Nsight Systems/Compute). Experience: team uses CUDA and Nsight Systems routinely. Adoption: migrate the Awave-3D build from clang to nvc++ and validate numerical parity against existing baselines. Nsight Compute provides kernel-level roofline analysis, now applied per spatial tile, to identify which tiles are memory-bandwidth-bound versus compute-bound and to objectively scope where a lower-precision tile actually pays off.

RTX 5090 (dedicated hardware, requested, 32 GB GDDR7 plus a server with 64 GB of RAM). This GPU is also based on the Blackwell architecture and shares with same-generation professional cards the same memory bandwidth (1,792 GB/s) and the same 5th-generation tensor cores, with FP8/FP4 support and negligible native FP64 throughput. For a memory-bandwidth-bound finite-difference solver like ours, it is memory bandwidth,

not raw CUDA-core count, that determines performance. This makes the RTX 5090 a representative development platform for P-AMR, at a fraction of the cost of an equivalent professional card and within the per-item equipment ceiling set by FAPESP's Regular Research Grant rules for this type of request. The absence of fast native FP64 is not a new disadvantage: it is exactly the precision regime that P-AMR already assumes and exploits, via Ozaki-scheme emulation and mixed-precision GEMM, so the scientific question this project answers, precisely which tiles need emulated FP64 and which do not, remains the same. Its 32 GB of GDDR7 (without ECC) support kernel iteration and the tuning of tagging criteria on representative subsets of the domain; as dedicated, queue-free hardware, it enables fast iterative debugging that a shared batch system is not well suited for. FP64 ground truth and production validation continue to be generated on Santos Dumont, with ECC-protected memory.

Santos Dumont (LNCC/SINAPAD, Tier-0 national supercomputer, Petrópolis-RJ). Provides the large-scale validation environment. Its BullSequana X1000 partition includes 94 nodes with $4 \times$ NVIDIA Volta V100 GPUs each, matching the hardware used for the original GPUZIP baselines, plus a dedicated AI node with $8 \times$ V100 16GB over NVLink. Its newer BullSequana XH3000 partition adds 62 nodes with $4 \times$ NVIDIA Hopper H100 SXM 80GB over NVLink, 36 NVIDIA Grace-Hopper GH200 superchip nodes, and 18 nodes with $2 \times$ AMD Instinct MI300A APUs (CDNA3 GPU cores, 128 GB HBM3 each). This heterogeneity lets us generate FP64 ground truth and run the full P-AMR criteria sweep at scale on NVIDIA hardware, and separately test whether the tagging methodology and block-tiling approach generalize to a non-NVIDIA architecture.

nvCOMP / Bitcomp (GPU-side lossy compressor). Experience: integrated and validated in GPUZIP v2.0, where Bitcomp delivered the best accuracy-compression balance on V100. Adoption here: the storage mechanism for differential checkpointing, now indexed by the same tile identifiers used for precision tagging, so each tile carries a single, jointly-tuned precision-and-compression error budget.

cuBLAS (mixed-precision extensions) and cuSOLVER (iterative refinement). Both ship inside the HPC SDK. Experience: team has used BLAS/LAPACK routines in prior HPC work. Adoption: applied only within tiles that P-AMR flags as containing dense sub-kernels needing accuracy beyond native low precision (FP32 compute with FP64 accumulation, and cuSOLVER iterative refinement).

Ozaki-scheme FP64 emulation via low-precision tensor cores. Adoption: reserved for the subset of tiles that P-AMR's sensitivity criteria mark as accuracy-critical (near the wave-front, high adjoint-sensitivity magnitude), rather than applied uniformly. This directly tests

whether selective emulation is cheaper than emulating everywhere while preserving image fidelity. Recent work has demonstrated up to $2.3\times$ speedup over native FP64 DGEMM on Blackwell GPUs [7], measured on the professional RTX PRO 6000 card; we expect comparable results on the RTX 5090, since both share the same tensor-core generation.

cuFFT. Experience: used in prior spectral operator work. Adoption: benchmarked as an alternative to FDM stencils for wavefield updates within selected, precision-tagged tiles, characterising the memory-bandwidth versus compute trade-off relative to the standard finite-difference path.

AMD Instinct MI300A (CDNA3, Santos Dumont, exploratory). As an additional validation activity, we assess how much of the P-AMR framework, that is, the block/tile decomposition, the four tagging criteria, and the joint precision-compression error budget, transfers to AMD’s ROCm/HIP ecosystem (rocBLAS, hipBLAS, and AMD’s own reduced-precision matrix paths), without assuming feature parity with the CUDA-specific Ozaki-scheme emulation or nvCOMP/Bitcomp compression path. The goal is to determine whether Precision-AMR is a portable methodology or an NVIDIA-specific optimization.

4 Evaluation

All experiments span three datasets (Large, Marmousi3D, Salt), three checkpointing algorithms (Revolve, zCut, Uniform), and the four P-AMR tagging criteria (individually and in combination). Kernel-level profiling uses Nsight Compute (roofline, memory-traffic breakdowns); timeline profiling uses Nsight Systems. Implementation is in C/C++ with CUDA under the HPC SDK, with OpenMP Cluster for multi-node scaling. Accuracy is evaluated on both the wavefield and the final RTM image, since a tiling that preserves wavefield PSNR/SSIM may still, in principle, shift the migrated image if error is spatially correlated with reflectors. For the cross-architecture activity, the same accuracy metrics are computed on Santos Dumont’s AMD MI300A partition and compared against the NVIDIA baselines, to characterise how much of the achieved speedup and accuracy trade-off is architecture-independent.

5 Project Support Details

We request a dedicated local server equipped with an NVIDIA RTX 5090 GPU (32 GB GDDR7) and 64 GB of RAM, with an estimated cost of up to R\$ 55,000 in Brazil, within the per-item equipment ceiling set by FAPESP’s Regular Research Grant rules. This GPU shares with same-generation professional Blackwell cards the same memory bandwidth

(1,792 GB/s) and the same 5th-generation tensor cores, which makes it representative for kernel development on a memory-bandwidth-bound solver like ours, at a fraction of the cost of an equivalent professional card. Its negligible native FP64 throughput makes it the scientifically relevant target for the selective-emulation study central to Phase 4: it forces the question of whether Precision-AMR’s selective use of Ozaki-scheme emulation can match full-domain emulation at a fraction of the cost. As dedicated, queue-free hardware, it also supports the iterative day-to-day development that a shared batch system is not well suited for. Large-scale validation, that is, the P-AMR criteria sweep across datasets and checkpointing algorithms, FP64 baseline generation, multi-GPU scaling, and the AMD MI300A portability test, runs separately on the Santos Dumont supercomputer (LNCC/SINAPAD), accessed through LNCC’s own allocation program and outside the scope of this request.

Resource Request

Resource	Request
Physical hardware	1× server with an NVIDIA RTX 5090 GPU (32 GB GDDR7) and 64 GB of RAM, dedicated local development hardware, used throughout the project (estimated cost: up to R\$ 55,000)

Compute time on Santos Dumont is requested separately through LNCC’s own project allocation program (SINAPAD) and is not part of this hardware request.

References

- [1] Brisa N. Davis and Randall J. LeVeque. Adjoint methods for guiding adaptive mesh refinement in wave propagation problems. *arXiv:1511.03645*, 2015.
- [2] Gabriel Fabien-Ouellet. Seismic modeling and inversion using half-precision floating-point numbers. *Geophysics*, 85(3):F65–F76, 2020.
- [3] Longfei Gao and Kevin Harms. Half precision wave simulation. *arXiv:2310.00236*, 2023. Revised 2024.
- [4] LSC-Unicamp. GPUZIP source code. <https://github.com/LSC-Unicamp/GPUZIP>, 2025. Accessed 2026.

- [5] Thiago Maltempi, Sandro Rigo, Marcio Pereira, Hervé Yviquel, Gustavo Leite, Otavio Lee, Jesse Costa, and Guido Araujo. Checkpointing fine-tuning for accelerating seismic applications in GPUs. *International Journal of High Performance Computing Applications*, 39(6):784–802, 2025.
- [6] Solvi Thrastarson, Martin van Driel, Lion Krischer, Christian Boehm, Michael Afanasiev, Dirk-Philip van Herwaarden, and Andreas Fichtner. Accelerating numerical wave propagation by wavefield adapted meshes. Part II: full-waveform inversion. *Geophysical Journal International*, 221(3):1591–1604, 2020.
- [7] Takumi Uchino et al. Guaranteed DGEMM accuracy while using reduced precision tensor cores through extensions of the Ozaki scheme. arXiv:2511.13778, 2025. Presented at SCA/HPCAsia 2026, Osaka, Japan.
- [8] Martin van Driel, Christian Boehm, Lion Krischer, and Michael Afanasiev. Accelerating numerical wave propagation using wavefield adapted meshes. Part I: forward and adjoint modelling. *Geophysical Journal International*, 221(3):1580–1590, 2020.
- [9] Xiujin Yang, Jianping Huang, Yi Shen, and Liang Chen. Elastic reverse time migration using an efficient and accurate adaptive variable grid discretization method. *Journal of Geophysics and Engineering*, 21(5):1463–1476, 2024.